

Part 1 of 2

Flexible Electronics: The Next Ubiquitous Platform

Just as the IC replaced discrete circuit board electronics, flexible electronics will almost inevitably, by virtue of the ever-demanding end-user, supersede solid-state ICs. So let us find out a bit more about flexible electronics, the materials required for their fabrication, the fabrication processes and the applications

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Flexible electronics is lightweight, rugged, bendable, roll-able, portable and potentially foldable. Ever-evolving advances in thin-film materials and devices have fuelled many of the developments in the field of flexible electronics. These advances have been complemented with the development of new integration processes, enabling wafer-scale processes to be combined with flexible substrates.

Diodes and transistors are two of the most common active thin-film devices used in digital and analogue circuits. While they have been successfully used in flexible platforms, their performance and applicability is limited by requirement of exotic device architectures and novel materials. In order to achieve the goal of full-system integration in next-generation flexible systems, a paradigm shift in design and fabrication is necessary.

There are many potential applications of flexible electronics in health-care, automotive, human-machine interfaces, mobile communications and computing platforms, and embedded systems in both living and hostile environments. Besides, there are market-specific applications, such as human-machine interactivity, energy storage and generation, mobile communications and networking, touching the application of flexible electronics on ubiquitous computing platforms throughout.

Flexibility in electronic materials is very attractive for medical and bio-

engineering. Living organisms are intrinsically flexible and malleable. Thus, flexibility is a necessity for successful integration of electronics in biological systems. Further, in order to carry out daily tasks, flexibility is less likely to hinder over stiffness.

Materials

The fundamental properties of thin-film materials, as well as the quality of various device interfaces, give rise to inherent limitations in device performance. Each element of the flex circuit must be able to consistently meet the demands placed upon it for the life of the product. In addition, the material must work reliably in concert with the

other elements of the circuit to assure ease of manufacture and reliability.

The materials and technologies behind flexible substrates are an important consideration for flexible electronics. Perhaps two of the main flexible substrate candidates are plastic and stainless steel. Although stainless steel is incompatible with standard deposition temperatures, it results in a substantially heavier system due to its higher mass density—a critical consideration for portable flexible technologies. Also, stainless steel is not particularly deformable, and is thus unsuitable, in particular, for wearable electronics. Plastic substrates are lighter and deformable alternatives.

Substrates need to be solvent-resistant, so that standard optical photolithography process can be used. Additional substrate requirements include low cost (allowing large area, mass production) and moisture resistance. Table I compares some of the more critical properties of some key plastic substrates.

One of the main challenges facing plastic as a next-generation substrate is the substantially reduced processing temperature window. The maximum fabrication temperature is related to the glass-transition temperature above which inelastic deformation takes place and the substrate no longer retains its

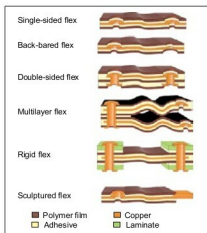


Fig. 1: Examples of basic and selected flexible circuit construction

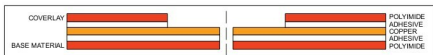


Fig. 2: Single-sided flex circuit